Question ID: 1efd7ef3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

What is the value of $\sin 42\pi$?

Answer

A. 0

B. $\frac{1}{2}$

C. $\frac{\sqrt{2}}{2}$

D. 1

Correct Answer: A

Rationale

Choice A is correct. The sine of a number t is the *y*-coordinate of the point arrived at by traveling a distance of t units counterclockwise around the unit circle from the starting point $(1, 0)$. Since the unit circle has a circumference of 2π units, it follows that one full rotation around the circle is equal to a distance of 2π units. Therefore, a distance of 42π units around the circle from the starting point $(1, 0)$ would result in exactly 21 full rotations, arriving back at the point $(1, 0)$. So, $\sin 42\pi$ is equal to the *y*-coordinate of the point $(1, 0)$, which is 0.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $\cos 42\pi$, not $\sin 42\pi$.